

Enhancing Water Transport Efficiency Through Geospatial Risk Modeling

Transforming multimodal open data into decision making

Candidate

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Master thesis

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*How can we define
the vulnerabilities of a
Water distribution
networks?*

Problem framing

Shifting the Dimensions of Risk

The Foundation (Sangroula et al., 2026)

1. Physical
2. Environmental
3. Operational

Our Approach

- The Challenge: Missing or inaccessible structural physical pipeline data.
- The Solution: Pivot the problem structure to exploit the vast availability of Open Data in the remaining two dimensions.
- Focus Area: Building a flexible risk framework driven entirely by Environmental and Operational indicators.



Context

*How can we define
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Problem framing

Shifting the Dimensions of Risk

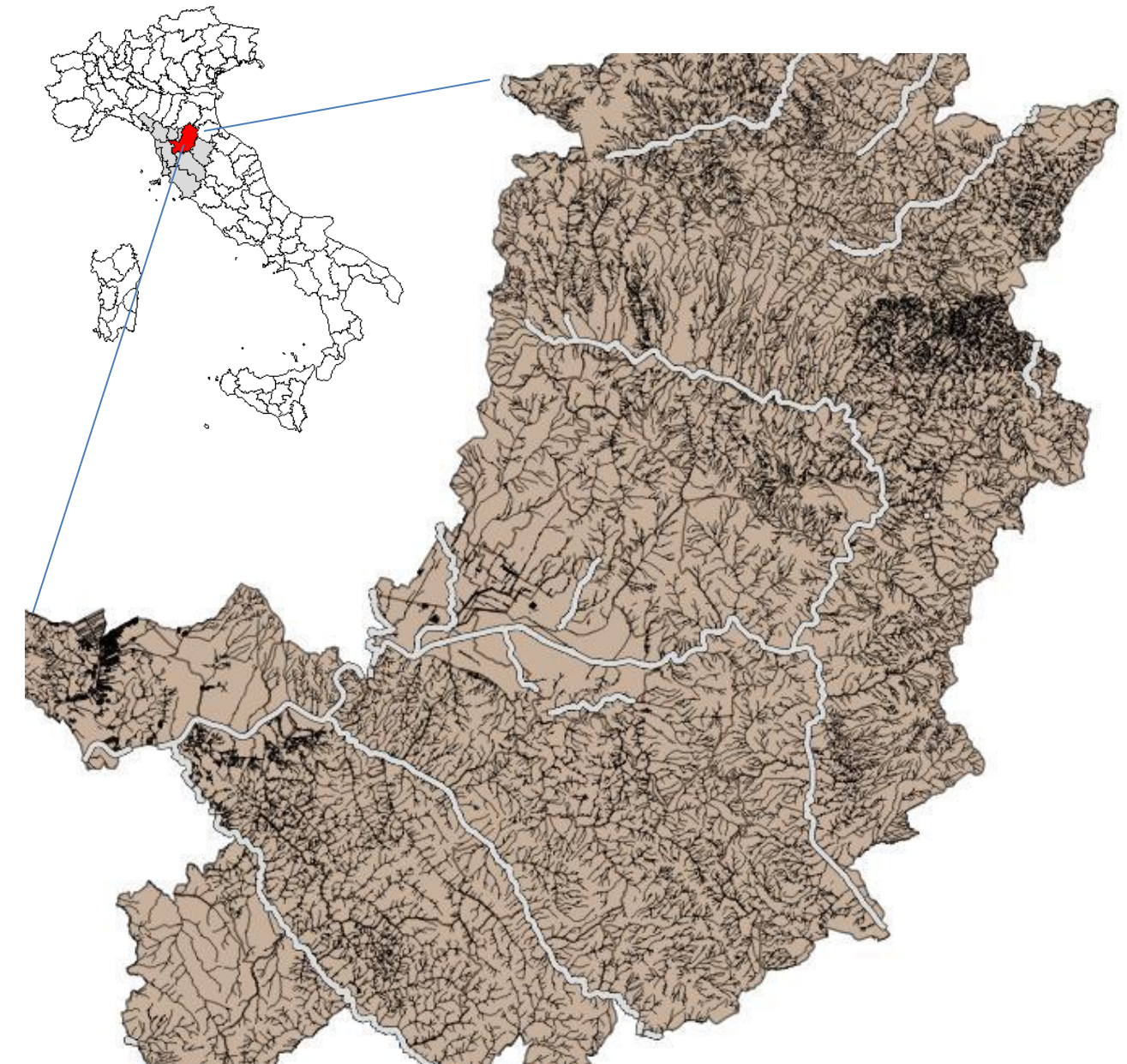
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Florence Province



Case study & geospatial scope

- Province of Florence (Firenze) initial phase.
- Initial scope manageable and controlled.
- Designed to scale the methodology across the Italian territory.
- Standardized Uber H3 grid
- Instead of mapping specific points we map zones
- If a pipeline is inside the zone, how hazardous would it be?

Research question & objectives

Main Research question

How can multimodal open data and patent data be used to reduce inefficiencies in water transportation systems through pipelines?



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Secondary objectives

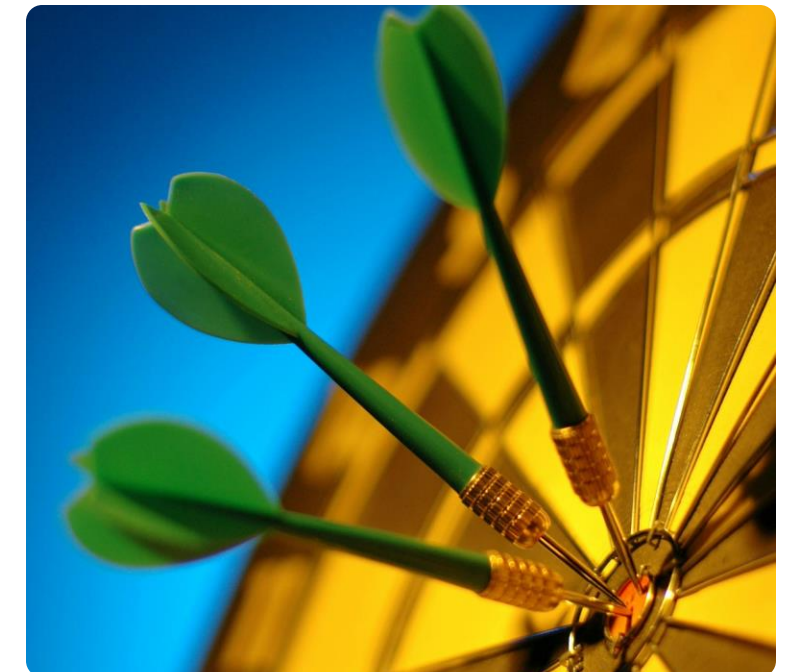
To identify and systematically catalog diverse repositories of multimodal open data and global patent databases relevant to the Italian water transport system.

Secondary objectives

To establish a comprehensive methodology for the integration and harmonization of heterogeneous datasets, ensuring interoperability between various knowledge fields and from structured to unstructured data.

Secondary objectives

To design a data-driven framework that utilizes integrated insights built on causes and effects to support the prioritization of infrastructure interventions and the optimization of resource allocation.





 Methodology
framing

*How can we pair causes
with solutions?*

Approach

Vulnerability layer & Mitigation layer

Vulnerability layer

- Compiles all collected spatial layers into a single, unified index for every individual H3 hexagon cell.
- Output a single risk score; and calculates Feature Importance metrics.
- It breaks down the exact percentage of why a specific hexagon is risky (e.g., 70% Slope, 20% Soil Type, 10% Rainfall).



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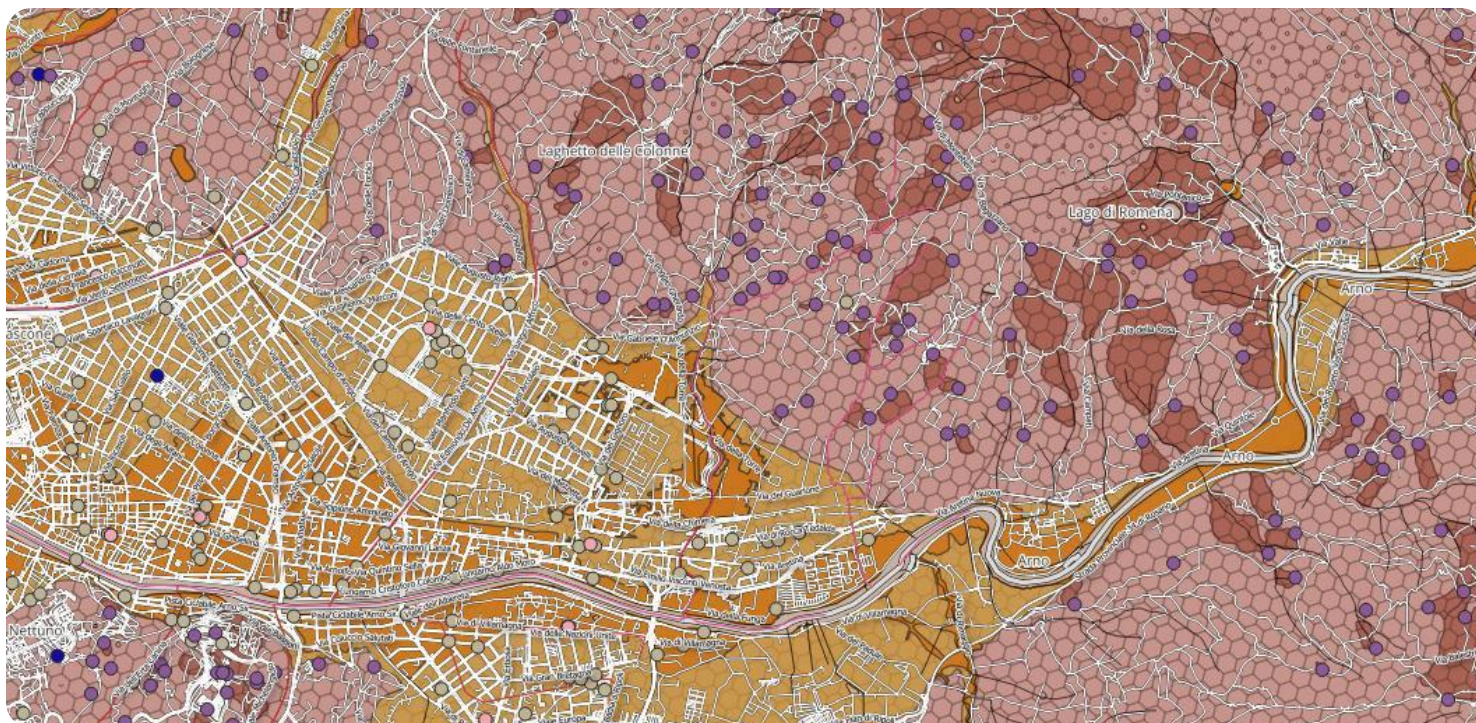
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Mitigation layer

- The system ingests the Risk Index and the Feature Importances generated by the vulnerability layer.
- Rather than giving generic warnings, it cross-references the hazard profile of the hexagon (e.g., high soil acidity or steep slope).
- The system automatically recommends specific **patented** engineering solutions tailored precisely to mitigate the risks of that area.



Data sources

Multimodal, reliable and open access

Environmental risk sites and Patent data

Programs like **SIR** from **Regione toscana**, **IdroGEO** from **Isprambiente**, **CFTI** **landslides** from the national institute of geophysics and vulcanology (**INGV**) and **Copernicus** crops data.

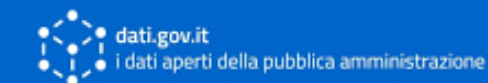
Patent data enhances technical knowledge about the physical factors that affect the pipeline systems at hand



Preliminary exploration EUROSTAT & ISTAT



Dati.gov.it and ARPA entities



Open data regional sites



Main sources by content

- Official National Hazard Inventories for landslides, floods, seismicity and wildfires
- Decades of localized rain , temperature, humidity and wind gust history
- Communal income and population census
- Pedological soil databases, sentinel-1 soil displacement points
- Streets, infrastructural assets, waterways (OSM)
- Pendolarismo, daily commuters between communes
- Natural disasters interventions
- High-resolution DTMs

Patents retrieval

Retrieval strategy

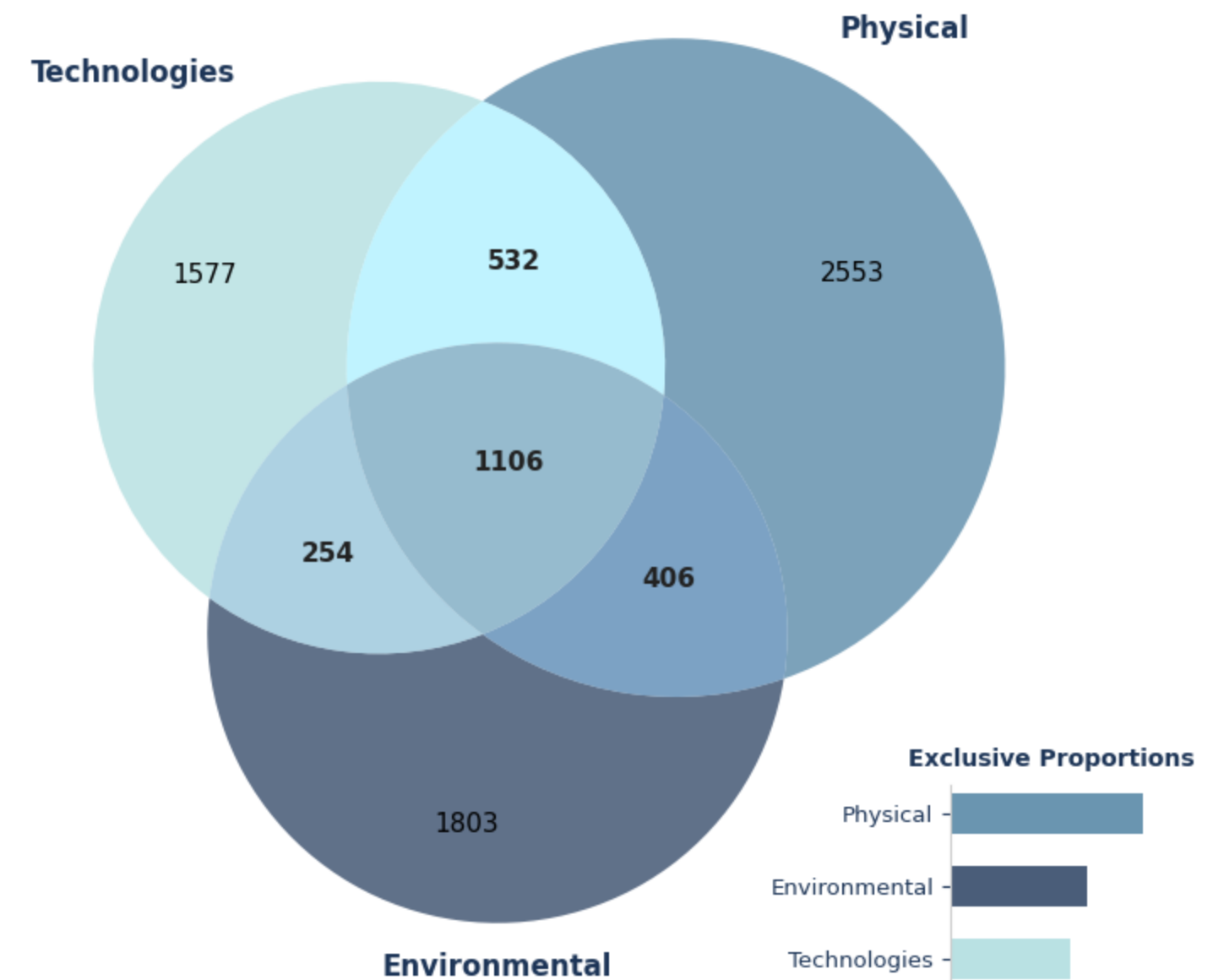
Extracted a comprehensive macro-pool of global patents targeted at:

- Water distribution networks
- Bulk water transportation
- Sewage and wastewater infrastructure

The classification dimensions

- **Physical:** Structural pipeline faults, joints, reinforcements, and physical assets.
- **Environmental:** Soil stabilization, erosion control, and external terrain-defense solutions.
- **Technologies:** Digital systems, monitoring tools, smart valves, and software layers.

Sub-topic(Dimensions) Distribution & Overlap Analysis



Methodology

Creating a framework for the risk assessment solutions

Data cube structure

Generate a star schema that allows the system to separate highly granular data from static data, thus allowing the risk assessment to be made with the contextual information of the static properties, plus the situational conditions that change in a daily, weekly or monthly basis



Causes

- Seismic/site intensity
- Rain
- Temperature anomalies
- Wind gust anomalies
- Wildfires (EFFIS)
- Pendolarismo daily commuters

Conditions

- DTM's
- Soil data, pedologia
- Idrografia
- Streets and water ways
- Infrastructure

Effects

- Interventions
- Economical data
- Population census
- Landslides/Floods history
- Sentinel1 soil displacement points

Conclusions

A risk model in it's early phases

Value of the proposal

- The project works on open data thus readily available for anybody to use, no private data (thus far)
- The model is highly scalable to cover the whole country, since the data is collected mostly with a remote sensing methodology
- The model could be used for other risk related fields in Italy aside from WDS, since it has a big component of environmental risk

Negatives of the proposal

- The model does not include insitu physical properties of the pipelines, reducing technical information but increasing flexibility
- The project is based on environmental, economical and social sources which can have low granularity
- The model thus far does not contain a label which serves as a control group for the risk assessment

Thank you for your
attention!

*By being in the initial phase, there's still
things to change and improve*

Any suggestions you might have
regarding the data: **collection**,
methodology structure or the
connection between **risk and**
patents are very welcome